

Six-Axis Force Sensor Serial Port Output Method

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1. Clarification

This document describes the protocol interaction. The module is in modbus protocol within 30s of power up. If there is no interaction with modbus protocol after 30s of power-up, the module will automatically use the customized protocol to output continuously. If there is no interaction with modbus protocol after 30s of power-up, the module will automatically use the customized protocol to output continuously.

2. Setting the Serial Output Mode-RTU

2.1 Communication notes

The transmitter adopts standard Modbus-RTU communication format, baud rate 115200bps/8bit/no parity bit. It supports Modbus-RTU slave mode, and the communication data follows the big end alignment mode.

Note: The so-called big-end mode, where the high byte of data is stored in the low address of the memory and the low byte of data is stored in the high address of the memory, is somewhat similar to treating data as a string sequence: the address is increased from small to large and the data is placed from high to low.

16-bit integer data e6 84 for 0xe684.

32-bit integer data is in the format 3412; 12 34 56 78 means: 0x56781234.

ModBus Node Address:

Module fixed station address numbers are: #1, #2, #3, #4, #5, #6

2.2 Modbus Command Form

ModBus RTU support for 03H, 06H, 10H functions					
Other PLC Addresses	Siemens PLC Address		Parameter description	Default Value	Data type Operation attributes
00/01	40001/40002		Display weight (ignore decimal point)		Int32 Read-only
02	40003	.0	0=dynamic; 1=steady state		bit
		.1	0 = non-zero bit; 1 = zero bit		bit
		.2	0 = gross weight mode; 1 = net weight mode		bit
		.3	0 = no upper overload; 1 = upper overload		bit

		.4	0 = no lower overload; 1 = lower overload		bit	
		.5 ~ 15			bit	
02	40003	Manipulate: 0: None; 1: Clear; 2: Tare; 3: Clear; 4: X10; 5: Gross net mode switching			Int16	Read
03	40004	Number of decimal places, optional: 0, 1, 2, 3, 4		0	Int16	Fill out or in
04/05	40005/40006	Gross weight (ignoring decimal points)			Int32	Read-only
06/07	40007/40008	Tare weight (ignoring decimal points)			Int32	Fill out or in
08	40009	Indexing value 1 (corresponding to high range), optional: 1, 2, 5, 10, 20, 50, 100.		1	Int16	Fill out or in
09	40010	Indexing value 2 (corresponding to low range), optional: 1, 2, 5, 10, 20, 50, 100.		1	Int16	Fill out or in
10/11	40011/40012	Full scale (high range)		6000	Int32	Fill out or in
12/13	40013/40014	Intermediate range (low range)		0	Int32	Fill out or in
14	40015	Units, optional: 0: kg; 1: g; 2: t		0 (kg)	Int16	Fill out or in
15	40016	Power-on zeroing range (0 to 100): 0% to 100%		2	Int16	Fill out or in
16	40017	Manual zeroing range (0 to 100): 0% to 100%		5	Int16	Fill out or in
17	40018	Zero tracking range (0 to 65535 ignoring decimal points)		10	Int16	Fill out or in
18	40019	Stabilized display range (0 ~ 99) : 0:OFF; 1:0.1d ... 98:9.8d; 99:9.9d.		3 (0.3d)	Int16	Fill out or in
19	40020	Filter level: (0 to 5) The larger the value, the heavier the filtering.		3	Int16	Fill out or in
20	40021	Calibration point serial number: (0 to 5) 0: represents the zero point; 1: represents the first calibration point		0	Int16	Fill out or in

21	40022	Calibration execution/calibration status: Write: 1: Execute calibration at this point, other values are invalid Read: Calibration countdown time, 0: Calibration complete; -1: calibration failed	0	Int16	Fill out or in
22/23	40023/40024	Calibrator weight value (weight of weight)	0	Int32	Fill out or in
24/25	40025/40026	Calibrated internal code values	0	Int32	Fill out or in
26/27	40027/40028	Total sensor range (sum of all sensor ranges)	1000.0	Int32	Fill out or in
28/29	40029/40030	Average sensor sensitivity, 200000 for 2.00000mV/V	200000	Int32	Fill out or in
30	40031	Slave address (own address): 1 to 32	32	Int16	Fill out or in
31	40032	Software version number:		Int16	Read-only
34/35	40035/40036	Device Serial Number		Int32	Read
96	40097	AD Sampling Rate Selection: 0:6 times/sec; 1:12 times/sec; 2:25 times/sec; 3:50 cycles/sec; 4:100 cycles/sec; 5:200 cycles/sec; 6:40 cycles/sec; 7:800 cycles/sec; 8:1600 cycles/sec	4	Int16	Fill out or in
98/99	40099/40100	AD internal code		Int32	Read

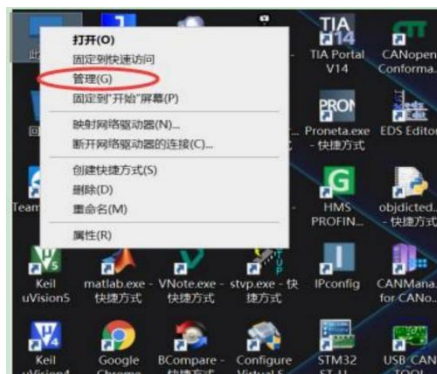
3. Setting the serial port parameters on the computer side

3.1 Preparation of the serial port driver on the computer side

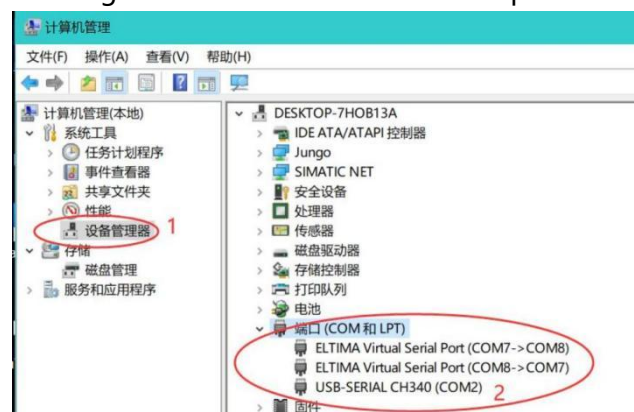
If the instrument is connected to the computer via RS485 converter. Make sure that the driver of the converter has been installed successfully and can communicate with the computer normally.

3.2 Viewing Serial Port Numbers

Right click on “My Computer” and select Manage.:



After double clicking “Device Manager”, double click “Ports (COM and LPT)” to enter the following interface to view the serial port number: COM2.



3.2 Configuring Serial Port Parameters

3.2.1 Using the Serial Debugging Assistant

When using the serial port debugging assistant to test the communication, you need to make sure that the parameter settings in the software are the same as those of the instrument. For example, the baud rate of the instrument is 115200, the communication mode is rtu, and the data format is 8-n-1.

The software settings are as follows: (Receive setting: Hex, Send setting: Hex)



3.2.2 Use of debugging software

You can use the matching debugging software to communicate with the module, matching debugging software provided by our company. The main interface of the software is shown below:



4. Serial Assistant Communication Test

Once the parameters are set, you can send commands to interact with the meter for weight data:

Meter Slave No. 01, Read Weight Data Command: 01 03 00 00 00 02 C4 0B

Meter Slave No. 02, Read Weight Data Command: 02 03 00 00 00 02 C4 38

Instrument Slave Number	03 , Read Weight Data Command: 03 03 00 00 00 02 C5 E9
Instrument Slave Number	04 , Read Weight Data Command: 04 03 00 00 00 02 C4 5E
Instrument Slave Number	05 , Read Weight Data Command: 05 03 00 00 00 02 C5 8F
Instrument Slave Number	06 , Read Weight Data Command: 06 03 00 00 00 02 C5 BC

5. Examples of read and write commands

Example of interacting with the meter using slave number 1:

The protocol for reading slave address 1 (01 03 00 00 00 00 02 C4 0B) is specified as follows:

从站地址	功能码	数据地址	数据长度	CRC校验
01	03	00 00	00 02	C4 0B

The protocol for writing slave address 32 (20 06 00 1E 00 01 2E BD) is specified as follows:

从站地址	功能码	数据地址	数据值	CRC校验
20	06	00 1E	00 01	2E BD

5.1 Examples of common parameter interactions

5.1.1 Read real-time value command: (protocol address 0000H/0001H, PLC address: 40001/40002)

Dispatch: 01 03 00 00 00 02 C4 0B

Return: 01 03 04 00 00 00 00 FA 33

According to the returned data, the current value of the meter is 0. Since the number of decimal places is 0, the corresponding number of the meter is 0;

Dispatch: 01 03 00 00 00 02 C4 0B

Return: 01 03 04 61 4E 00 BC 84 69

According to the returned data, the current value of the meter is 0x00BC614E = 12345678, since the number of decimal places is 0, the corresponding number of the meter is 12345678;

5.1.2 Read real-time status command: (protocol address is 0002H, PLC address is: 40003)

Dispatch: 01 03 00 02 00 01 25 CA

Return: 01 03 02 00 20 B9 9C

According to the returned data, the current value of the instrument is 0x0020; it means dynamic, non-zero position, gross weight mode, no upper overload, no

lower overload, single range, normal display weight, in weighing operation mode.

The significance of the bits corresponding to the returned data is as follows:

.0	0=dynamic; 1=steady state
.1	0 = non-zero bit; 1 = zero bit
.2	0 = gross weight mode; 1 = net weight mode
.3	0 = no upper overload; 1 = upper overload
.4	0 = no lower overload; 1 = lower overload
.5	0 = in low range; 1 = in high range (often 1 for single range)
.6	0=normal display weight; 1=index value reduced by 10 times display weight
.7~15	reserve

5.1.3 Write clear instruction: (protocol address is 0002H, PLC address is: 40003)

Dispatch: 01 06 00 02 00 01 E9 CA

Return: 01 06 00 02 00 01 E9 CA

According to the returned data, the operation is successful.

5.1.4 Read decimal point instruction: (protocol address is 0003H, PLC address is: 40004)

Dispatch: 01 03 00 03 00 01 74 0A

Return: 01 03 02 00 00 B844

Indicates that the decimal point of the meter is 0, based on the returned data.

5.1.5 Write Decimal Point Instruction: (Protocol address is 0003H, PLC address is: 40004)

Dispatch: 01 06 00 03 00 03 39 CB

Return: 01 06 00 03 00 03 39 CB

Modify the decimal point of the meter to 3 digits.

5.1.6 Read Meter Slave Number Command: (protocol address 001EH, PLC address: 40031)

Dispatch: 01 03 00 1E 00 01 E4 0C

Return: 01 03 02 00 20 B9 9C

According to the returned data, it indicates that the slave number of the meter is $0x0020 = 32$.

5.1.7 Write Instrument Slave Number Command: (protocol address 001EH, PLC address: 40031)

Dispatch: 01 06 00 1E 00 01 28 0C

Return: 01 06 00 1E 00 01 28 0C

Based on the returned data, it indicates that the slave number of the meter is modified to 01.

5.2 Example of basic parameter operation

5.2.1 Read index instruction: (protocol address is 0008H, PLC address is 40009)

Dispatch: 01 03 00 08 00 01 05 C8

Return: 01 03 02 00 01 79 84

According to the returned data, it indicates that the meter has a dividing value of 1.

5.2.2 Modify indexing value instruction: (protocol address is 0008H, PLC address is 40009)

Dispatch: 01 06 00 08 00 05 C8 0B

Return: 01 06 00 08 00 05 C8 0B

Modify the scale value of the meter to 5.

5.2.3 Read Full Scale Command: (protocol address is 000AH/000BH, PLC address is 40011/40012)

Dispatch: 01 03 00 0A 00 02 E4 09

Return: 01 03 04 86 A0 00 01 12 99

The data is: 0x000186A0 = 100000, indicating a full scale of 100000.

5.2.4 Modify full scale command: (protocol address is 000AH/000BH, PLC address is: 40011/40012)

Dispatch: 01 10 00 0A 00 02 04 0D 40 00 03 30 A9

Return: 01 10 00 0A 00 02 61 CA

The data is: 0x00030D40 = 200000, indicating a full scale of 200000.

5.2.5 Read power-on clear range instruction: (protocol address is 000FH, PLC address is: 40016)

Dispatch: 01 03 00 0F 00 01 B4 09

Return: 01 03 02 00 14 B8 4B

The data is: 0x0014 = 20, indicating that the clearing range is 20% of full scale.

5.2.6 Modify the power-on clear range instruction: (protocol address is 000FH, PLC address is: 40016)

Dispatch: 01 06 00 0F 00 0A 39 CE

Return: 01 06 00 0F 00 0A 39 CE

The data is: 0x000A=10, which means that it is modified to 10%.

5.2.7 Read manual zero range: (protocol address 0010H, PLC address: 40017)

Dispatch: 01 03 00 10 00 01 85 CF

Return: 01 03 02 00 14 B8 4B

The data is: 0x0014 = 20, indicating that the clearing range is 20% of full scale.

5.2.8 Modify the manual clear range: (protocol address 0010H, PLC address: 40017)

Dispatch: 01 06 00 10 00 0A 08 08

Return: 01 06 00 10 00 0A 08 08

The data is: 0x000A= 10 , indicating that the clearing range is 10% of full scale.

5.2.9 Read Zero Tracking Range command: (protocol address 0011H, PLC address: 40018)

Dispatch: 01 03 00 11 00 01 D4 0F

Return: 01 03 02 00 14 B8 4B

5.2.10 Modify Zero Tracking Range Command: (protocol address is 0011H, PLC address is 40018)

Dispatch: 01 03 00 11 00 01 D4 0F

Return: 01 03 00 11 00 01 D4 0F

5.2.11 Read Stability Level command: (protocol address 0012H, PLC address: 40019)

Dispatch: 01 03 00 12 00 01 24 0F

Return: 01 03 02 00 0A 38 43

The data is: 0x000A= 10

5.2.12 Modify Stability Level command: (protocol address 0012H, PLC address: 40019)

Dispatch: 01 06 00 12 00 14 2F 71

Return: 01 06 00 12 00 14 2F 71

The data is: 0x0014= 20

5.2.13 Read Filter Level command: (protocol address 0013H, PLC address: 40020)

Dispatch: 01 03 00 13 00 01 75 CF

Return: 01 03 02 00 05 78 47

The data is: 0x0005= 5.

5.2.14 Modify Filter Level Command: (protocol address 0013H, PLC address: 40020)

Dispatch: 01 06 00 13 00 02 F9 CE

Return: 01 06 00 13 00 02 F9 CE

The data is: 0x0002= 2.

5.2.15 Modify Low Side Comparison Value Instruction: (protocol address is 0074H, PLC address is: 40117/40118)

Dispatch: 01 10 00 74 00 02 04 03 E8 00 00 75 08

Return: 01 10 00 74 00 02 01 D2

The data is: 0x000003E8= 1000.

5.2.16 Modify High-End Comparison Value Instruction: (protocol address 0076H, PLC address: 40119/40120)

Dispatch: 01 10 00 76 00 02 04 07 D0 00 00 74 2C

Return: 01 10 00 76 00 02 A0 12

The data is: 0x000007D0= 2000.

5.3 Example of calibration operation

5.3.1 Calibration Zero:

Select calibration point (protocol address 0014H, PLC address: 40021):

01 06 00 14 00 00 C9 CE

Execute calibration point (protocol address 0015H, PLC address: 40022)

: 01 06 00 15 00 01 59 CE

Select Calibration Point 1 (Address 0014H, PLC Address: 40021)

: 01 06 00 14 00 01 08 0E

5.3.2 Write calibration weight (protocol address 0016H, PLC address: 40023/40024)

Load the actual weight (weight 1000) and write the loaded weight value 1000.

Dispatch: 01 10 00 16 00 02 04 03 E8 00 00 F2 F9

5.3.3 Calibration range

Select calibration 1 point (protocol address 0014H, PLC address: 40021): 01

06 00 14 00 01 08 0E

Execute calibration point (protocol address 0015H, PLC address: 40022)

: 01 06 00 15 00 01 59 CE

6. Continuous Output Protocol Description

This protocol is based on the transmission of hexadecimal numbers, and all packets are transmitted in small-end mode (low byte first) and output continuously.

6.1 Protocol Definitions

6.1.1 Communication protocol specification

baud: 460800

data bit: 8 bit

stop bit: one

Validity: no parity

6.1.2 Communication Protocol Format

No.	define	length	element	note
1	protocol header	2 byte	0xAA 0x55	fixed value

2	address	1 byte	0xFF	fixed value
3	data frame length	2 byte		Length of the entire message
4	command number	1 byte	0x10	fixed value
5	Channel 1 Data	4 byte	0x9C 0xFF 0xFF 0xFF	0xFFFFFFFF9C= -100
	Channel 2 Data	4 byte	0x64 0x00 0x00 0x00	Expressing 100
	Channel 3 Data	4 byte		
	Channel 4 Data	4 byte		
	Channel 5 Data	4 byte		
	Channel 6 Data	4 byte		
	Channel 1 Data	1 byte		
	Channel 2 Data	1 byte		
	Channel 3 Data	1 byte		
	Channel 4 Data	1 byte		
	Channel 5 Data	1 byte		
	Channel 6 Data	1 byte		
6	calibration	2 byte	CRC	CRC calibration

6.1.3 Channel Status Description:

		Status Description	Name
status value	0	0: Normal; 1: sensor disconnection alarm	Sensor connection status
	1	0: Chip working normally; 1: Chip failure	Chip Status
	2	0: unstable weight; 1: stable weight	weighing state
	3	0: non-zero bit; 1: zero bit	weighing state
	4	0: Normal; 1: Overload	weighing state
	5	0:Normal; 1:Failure to clear zero on power-up;	weighing state
	6-15	reserve	reserve